Lantern Kernel

The programmer's guide: writing TI-99/4A applications in assembly

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For a TI-99/4A with an F18A video card and a SAMS memory card
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Contents

0. What you get, what you need

1. Hello, kernel

2. The machine, in one page

3. The anatomy of a module

4. Calling conventions and registers

5. The screen

6. The keyboard

7. Memory

8. Several modules

9. Files

10. Time

11. Quitting

12. Building

13. Testing

14. The checklist: what has already cost a day

15. What the kernel does not do (version 1.1)

Appendix: a second module in five minutes

How to write an application for the TI-99/4A in TMS9900 assembly on the Lantern Kernel. The guide is a tutorial first (parts 1-3 get a program on the screen), then a tour of each service (parts 4-11), then the practical matter of building and testing (parts 12-13), and ends with a checklist of the mistakes that have already cost someone a day (part 14). Look up a single routine in ROUTINES.md; the machine's facts are in HARDWARE.md; how the kernel works inside is in ARCHITECTURE.md.

The kernel's source is the one place that is never out of date. Where this guide and a header comment in `kernel/` disagree, the comment is right and this guide should be fixed.

0. What you get, what you need

The machine the kernel assumes: a TI-99/4A with an **F18A** (or PICO9918) video card, a **SAMS** memory card of at least the size your application asks for (128 pages, 512K, by default), and a cartridge slot holding a **FinalGROM 99** (or any cartridge that presents 8K banks selected by writes to `>6000 + 2n`: the "8" image format; js99er and Classic99 load the same file). The loader refuses to start, with a plain message, if the card, its size or the F18A is missing.

What the kernel gives your program:

Screen	boots in 80 x 30 text (F18A "Text 80", a colour per character cell; 256 six-by-eight glyphs; 15 colour *roles* and five colour schemes; DMA row moves); `VMODE` sets **any F18A mode**: Text 80/40, Graphics I and II, Multicolor, enhanced colour, sprites, tile layer 2, the bitmap layer (part 5 and VIDEO-MODES.md)
Keyboard	64-key buffer, auto-repeat, SHIFT / CTRL / FCTN / ALPHA LOCK chords decoded to one byte each
Memory	7K of RAM that is always there (slot A), 8K of your code in the window at `>C000`, and up to 1M of SAMS pages you map into `>E000-FFFF`; a page pool to hand them out
Structure	a main loop that calls your module once a frame; up to dozens of 8K modules; `GOMOD` to switch, `MODCALL` to call into another module and come back
Files	open, read, write, close, delete through any DSR (disk controller, TIPI, ...)
Safety	the loader's checks, a clean power-up state, a clean exit to the TI title screen

What you need on the PC: Python 3, and xdt99 (`xas99.py` is the assembler). Put xdt99 beside this directory (`../xdt99`), or set `XAS99` to the path of `xas99.py`. A web browser with js99er.net for running; the headless simulator in `tools/` for testing.

1. Hello, kernel

Build and run the smallest example:

```
python3 tools/lkbuild.py examples/hello
python3 tools/sim9900.py build/hello/hello8.bin --keys "<ALPHA>hello" --frames 200
```

The first command assembles the kernel, the loader and your module and writes `build/hello/hello8.bin`, a cartridge image. The second runs it in the simulator and prints the screen as text. To run it in js99er, load `hello8.bin` as a FinalGROM/cartridge image with the F18A and a SAMS card (512K or more) switched on in its settings.

Start your own application with the scaffolder, which writes a working one-module program:

```
python3 tools/newapp.py myapp --name "MY APP"
python3 tools/lkbuild.py myapp
```

Anatomy of `examples/hello/hello.a99`

```
       COPY "kernel.equ"      the kernel's symbols (routines, variables, equates)
       COPY "macros.a99"      .enter, .leave, .push, .pop, .callmod
```

Every module begins with these two lines. `kernel.equ` is written by the build when it assembles the kernel: it holds the address of every kernel routine (`PUTS`, `KEYGET`, ...) and every equate (`COLS`, `RTEXT`, `APPRAM`, `KQUIT`, ...). Your module is assembled separately, loaded into SAMS pages, and never linked to the kernel: these symbols are the whole connection.

```
ATEXT  EQU  (RTEXT*16+RBG)*256        text on the page background
```

A cell's **attribute** is a byte: foreground colour *role* in the high nibble, background role in the low (part 5). It is passed in

the *high byte* of a register, as every byte value in this kernel is, so the equate multiplies by 256.

```
       AORG >C000
       DATA INIT,TICK,REDRAW,0        the entry table
```

A module is assembled to run at `>C000` and starts with its **entry table**: four words, the addresses of `INIT`, `TICK`, `REDRAW` and a reserved word (part 3).

```
INIT   .enter
       BL   @REDRAW
       .leave
       RT
```

`INIT` is called when the kernel switches to your module. `.enter` and `.leave` save and restore the return address on the software stack: any routine that calls another needs them (part 4).

```
TICK   .enter
       MOV  R0,R0
       JEQ  TK9                       no key
       CI   R0,KQUIT*256              FCTN = ?
       ...
       SETO @QUITF                    ask the kernel to quit
```

`TICK` is called **once every frame** (sixtieth of a second) with `R0` holding the next key typed, in the high byte, or `0` for none. Setting `QUITF` non-zero asks the kernel to quit to the TI title screen. A key's code is a byte: ASCII for printable keys, with FCTN and CTRL chords in `>C0-FF` and `>80-BF` (part 6).

That is a whole program. Everything else in this guide is about doing more in `INIT` and `TICK`.

2. The machine, in one page

```
CPU address space              SAMS page(s)        what lives there
>0000-1FFF  console ROM        (not mapped)
>2000-3FFF  SLOT A             >02-03, fixed       kernel + application variables, the stack
>4000-5FFF  DSRs, mapper regs  (not RAM)
>6000-7FFF  cartridge          (not mapped)        only the loader looks at it
>8300-83FF  scratchpad         (not mapped)        fast RAM: workspace, kernel hot variables
>8400-9FFF  I/O ports          (not RAM)
>A000-BFFF  THE KERNEL         >0A-0B, fixed       3.6K of 8K used
>C000-DFFF  SLOT C             your module (2 pages) the running module, 8K
>E000-FFFF  SLOT B             any 2 pages         your data: window >E000 and window >F000
```

- **Slot A** is where your *variables* live, always mapped: `APPRAM` (`>2300`) up to `APPTOP` (`>3F00`) is yours, 7,168 bytes; `MODRAM` (`>2200`, 256 bytes) is scratch for the running module and is not kept across a module switch. The stack is `>3F00-3FFF`.
- **Slot C** is where your *code* runs. Only one module is mapped at a time.
- **Slot B** is for *data* too big for slot A: you map one SAMS page at `>E000` and another at `>F000` as you need them (part 7).
- **VRAM** (16K, on the F18A): the screen's tables take `>0000-215F`; most of the rest is yours (part 5).

Register use, in one sentence: **R0-R9 are scratch, R10 is the stack pointer, R11 the return address, R12 the CRU base, R13-R15 are best left alone.**

3. The anatomy of a module

A module is up to 8,192 bytes of code and constant data assembled at `>C000`. The first four words:

Word	Name	Called	R0	R1
`>C000`	`INIT`	when the kernel switches to the module (boot, or `GOMOD`)	the page of the module before, 0 at boot	the argument `GOMOD` was given, 0 at boot
`>C002`	`TICK`	every frame	the next key code (high byte), or 0	
`>C004`	`REDRAW`	**never by the kernel**; by convention, your module's "draw the whole screen"		
`>C006`	reserved			

`>C008...`	your entries	by `MODCALL` from another module		

INIT sets the module up and draws. It runs with the screen already in Text 80 and colour scheme 0 (or whatever the previous module left), and with whatever the previous module drew still on the screen: clear it (`CLS`) if you do not want it. Slot A variables are as the previous module left them: *only the first module's `INIT` can count on zeros*; later `INIT`s must initialise what they use.

TICK does the work. It must return reasonably quickly (part 10): while it runs the keyboard is not polled, the sound list does not advance and the screen does not change. A `TICK` that has nothing to do should simply return. Setting `QUITF` ends the program after `TICK` returns; so does `INIT`. `GOMOD` (part 8) likewise takes effect after the routine that called it returns.

REDRAW is yours to use. The usual pattern is that `INIT` ends with `BL @REDRAW`, and other modules call it (through `MODCALL`) to put your screen back after a pop-up covered it.

A module may be **up to 8K**, but not a byte past it: the build stops with an error. Anything that does not fit is a second module (part 8), or data in a SAMS page (part 7).

The kernel needs nothing else from a module. Everything not in the entry table is yours to name, except for the names `kernel.equ` exports: the **public routines** (`kernel/public.txt`, documented in ROUTINES.md) and every **equate** in `kdefs.a99`. A clash with one of those is an assembly error ("duplicate symbol"): easy to see. The kernel's other labels are private and never reach your module, so a new kernel release cannot collide with your labels.

4. Calling conventions and registers

The stack. `R10` is a software stack that grows down from `STKTOP` (`>4000`) in slot A, 256 bytes (128 words). Only four macros touch it:

Macro	Does
`.enter`	push `R11` (the return address)
`.leave`	pop `R11` (then you `RT`)
`.push Rn`	push a register
`.pop Rn`	pop it back

A routine that calls another (`BL`) must begin with `.enter` and end with `.leave` then `RT`, because `BL` overwrites `R11`. A **leaf** (a routine that calls nothing) needs neither and may simply `RT`. A leaf may also *tail-call* with `B @OTHER`: `OTHER`'s `RT` then returns to the leaf's caller.

Flags do not survive `.leave` (it is a `MOV`, which sets the flags from `R11`). A routine that ends with `.leave` returns its result in a register; the caller tests that register (`MOV R0,R0` then `JEQ`). Leaves may return results in the flags, but it is simpler not to.

Registers.

Register	Rule
R0-R9	scratch. *Every routine's header says which it changes ("Uses R0-R7").*
R10	the stack pointer: only the macros change it
R11	the return address; `BL` overwrites it
R12	the CRU base: every user of the CRU loads it first; nothing relies on it across a call
R13-R15	the TMS9900 context-switch registers in the console's interrupt routine, and the GPL workspace's in DSRs: leave them alone unless you know why

The rule that matters most: a value you still need after a call goes in a register the callee does not use, or in memory. Read the callee's "Uses". The common trap: `PUTS` changes `R0-R7`; `DECW` changes `R0-R8`; a loop counter kept in `R7` across a `PUTS` is gone. Loop state belongs in `R8-R9` (with `DECW`, only `R9`) or in a slot A variable.

`R0` can never be an index register (`MOV @TABLE(R0),R1` does not assemble). `MOVB` sets only the high byte: after `MOVB @X,R0` the low byte is whatever it was, so `CLR R0` first if you then compare the whole word. `SLA Rn,0` shifts by `R0`'s low four bits, not zero. `xas99` evaluates expressions strictly left to right (`A+2*3` is `(A+2)*3`): parenthesise.

5. The screen

The screen is 80 columns by 30 rows. A **position** is `row * 80 + column` (0-2399). Each cell has a **glyph** (a byte, the name table at VRAM `>0000`) and an **attribute** (a byte, VRAM `>1800`): the foreground role in the high nibble, the background role in the low. Every routine that writes text takes the attribute in the high byte of a register, and glyphs likewise.

Colour roles and schemes

Palette entries 1-15 are *roles*, not colours (`kdefs.a99`):

Role	Name	Role	Name
1	`RBG` page background	9	`RSELBG` selection background
2	`RTEXT` text	10	`RSELFG` selection foreground
3	`RBOLD` emphasis	11	`RBARBG` status bar background
4	`RITAL` italic	12	`RBARFG` status bar foreground
5	`RBOLIT` bold italic	13	`RRULER` rules and frames
6	`RSCRPT` super/subscript	14	`RMARK` markers, labels
7	`RDIM` dim	15	`RACCNT` accent: the cursor
8	`RERROR` error		

An attribute names two roles: `(RTEXT*16+RBG)*256` is text on the page. Entry 0 is transparent (the backdrop shows) so attributes use 1-15 only. A **colour scheme** is 16 palette words: `SETSCH` (R0 = 0-4) loads one, and the whole screen changes colour without a cell being rewritten. The five built in are TI-Writer's Screen Color order: blue/white (default), green/black, green/white, light blue/black, black/white. Your own schemes: part 12.

Think of roles as a theme: say what a cell *means* (error, cursor, status bar), not what colour it is. The roles are named for a text editor, but nothing in the kernel depends on the names; use them as you like.

Writing text

Routine	Does
`PUTS`	R0 = position, R1 = a *length-prefixed* string, R2 = attribute. Uses R0-R7
`PUTC`	R0 = position, R1 = glyph (high byte), R2 = attribute
`FILLN`	R0 = position, R1 = glyph (high byte), R2 = count: glyphs only
`FILLA`	R0 = position, R1 = attribute (high byte), R2 = count: attributes only
`DECW`	R0 = position, R1 = unsigned number, R2 = attribute; returns R3 = digits
`HEX2` / `HEX4`	R0 = position, R1 = number, R2 = attribute: two / four hex digits
`CLS`	R1 = attribute: every cell a space in it
`BOX`	R0 = top-left position, R1 = interior width, R2 = interior height, R3 = attribute: a frame

Strings are length-prefixed: `STRI 'text'` writes the length byte then the characters, so `LI R1,TITLE` / `BL @PUTS` just works. If you write them by hand with `BYTE n` then `TEXT`, count carefully (a miscounted string is the commonest bug) and put `EVEN` after any byte data before the next code label, or the label lands on an odd address where the TMS9900 silently executes from the even address below.

Positions past the end of a row simply continue on the next; nothing clips.

Glyphs

The pattern table holds 256 glyphs of 6 x 8 pixels (Text 80 shows the left six bits of each byte): UI glyphs 0-31 (box lines, arrows, markers, shades), ASCII 32-126, 127 the "Latin-1" marker, **underlined ASCII 128-222 (code + 96)**, more UI glyphs 223-255. The chart is GLYPHS.md; `examples/diag` draws it on screen (`>00..>E0`). To underline a character, add 96 to its code. The font is generated by `tools/font.py`; to use another, give your application its own `databank.a99` (part 12).

VRAM

The standard map, which every mode but Graphics II shares (VIDEO-MODES.md part 3 has the details and Graphics II's

map):

Range	Holds
`>0000-095F`	name table (2,400 bytes at 80 x 30)
`>0960-097F`	the GPU's mailbox and program (`GPUMB`, `GPUAD`)
`>09C0-09DF`	Graphics I's colour table (`GM1CT`)
`>0A00-135F`	the attribute table (`ATRTAB`, 2,400 bytes at 80 x 30)
`>1400-14FF`	sprite attribute table (`SATAB`)
`>1500-17FF`	**free** (768 bytes)
`>1800-1FFF`	pattern table, plane 0: the 256 glyphs (`PATTAB`)
`>2000-2FFF`	**free**, or the extra pattern planes of enhanced colour 2 and 3
`>3000-35FF`	**free**, or the sprite patterns (`SPTAB`)
`>3600-36FF`	the file service's record buffer
`>3700-37D7`	the file service's PAB
`>37D8-3FFF`	the disk controller's own: **never touch**

Write to VRAM with `VMBW` (block), `VFILL`, `VSBW` (one byte) and read with `VMBR`, `VSBR`. Each sets the VDP address and then moves bytes; the address auto-increments. **Interrupts are masked (`LIMI 0`) everywhere except inside `VSYNC`**, because an interrupt between the two bytes of an address write corrupts it; your code, in `TICK`, always runs with interrupts masked, and you leave it that way.

Moving rows: `VSHIFT`

`VSHIFT` copies R2 rows of `TXCOLS` bytes (80 in the boot mode) from VRAM address R0 to R1 using the F18A's DMA engine, started by a tiny GPU program the kernel installs. Scrolling a region of the screen is two `VSHIFT`s (name table and attribute table) and a redraw of the exposed row. R3 = 0 copies forwards, 2 backwards (for overlapping moves; then R0 and R1 point at the *last* byte of each block). It waits for the GPU to finish. Not every F18A firmware has the DMA engine: the kernel tests it at boot and sets `NODMA` non-zero when it does not; code that scrolls should redraw rows itself in that case.

Other display modes

`VMODE` switches the F18A to any of its modes: R0 = the base mode (`VMT80`, `VMT40`, `VMGM1` Graphics I, `VMGM2` Graphics II, `VMMC` Multicolor), R1 = the options added together (`VO30` 30 rows, `VOPOS` a colour attribute per position, `VOTE1-3` / `VOSE1-3` enhanced colour for tiles / sprites, `VOSPR` sprites on, `VOS16`, `VOSMG`):

```
        LI   R0,VMGM1              Graphics I, 8-colour tiles (enhanced colour 3), an attribute per position
        LI   R1,VOPOS+VOTE3
        BL   @VMODE
```

It re-initialises the registers and the VRAM tables (a blank screen, the glyphs, the colour scheme) and sets `TXCOLS`, `TXROWS`, `TXCELLS`, `TXATTR` and `NAMBAS`, which the text routines follow; **everything on the screen is gone afterwards**. `VREG` (R0 = register, R1 = value in the high byte) sets the rest: tile layer 2, scrolling, the bitmap layer, the sprite limits. The text routines work in Text 80, Text 40 and Graphics I; in Graphics II, Multicolor and the sprite and layer work you write the VRAM yourself. Two things trip people up: in an enhanced colour mode **the attribute byte is flags** (priority, flips, transparency, palette select), not a pair of colour roles; and **Graphics II uses the VRAM the file service needs**, so do not read or write files while it shows. `examples/modes` has a screen for every mode and VIDEO-MODES.md explains each, with pictures.

The cursor

There is no hardware cursor: draw one by giving a cell the accent attribute `(RBG*16+RACCNT)*256` (`ACUR` in `diag.a99`) and putting the old cell's attribute back when it moves.

6. The keyboard

A **key code** is one byte. The kernel decodes the whole chord (key + modifiers) to a single code, so your program never tracks SHIFT or CTRL itself:

Codes	Meaning
`>20-7E`	printable characters, as typed (SHIFT and the characters FCTN shows are included)
`>0D`	ENTER (with any modifier but CTRL)
`>80-BF`	CTRL + key: `>80 OR (the key's code AND >3F)`: CTRL A = `>81` ... CTRL Z = `>9A`, CTRL 0-9 = `>B0-B9`, CTRL ENTER = `>8D`
`>C0-FF`	FCTN + a key that has no character of its own: FCTN E/S/D/X (the arrows) = `KUP KLEFT KRIGHT KDOWN` = `>C5 >D3 >C4 >D8`; FCTN 1-9 = `>F1-F9` (DEL, INS, ERASE, CLEAR, BEGIN, PROC'D, AID, REDO, BACK; `KDEL KINS ...` in `kdefs.a99`); FCTN 0 = `>F0`; FCTN = = `KQUIT` = `>FD`

FCTN takes precedence over CTRL over SHIFT. ALPHA LOCK capitalises letters typed with no modifier. The complete table is in HARDWARE.md.

A code is produced **when a key goes down**, with the modifiers held then; releasing a modifier before the key changes nothing; holding a key repeats it after 24 frames and then every 4 (400 ms, then 15 a second). Codes go into a 64-entry buffer.

Getting keys. `TICK` is handed one key per frame, in `R0` (high byte). To take more (a batch of typing), call `KEYGET` yourself: it returns the next code in `R0`, or 0. To wait for a key in a moment that really is modal ("Press any key"), `KWAIT` runs frames until one arrives; `KFLUSH` throws away what is buffered. Compare with high-byte constants: `CI R0,KQUIT*256`, `CI R0,>2000`, `CI R0,>9400` (CTRL T).

Modifiers separately. `KMODS` holds the modifiers of the last code (`>80` FCTN, `>40` SHIFT, `>20` CTRL, `>10` ALPHA LOCK) in the high byte, and `ALOCK` is non-zero while ALPHA LOCK is down.

The console's own QUIT (FCTN =) is *disabled*: the kernel sets the console interrupt routine's flag byte to `>50`, so FCTN = reaches your program as code `KQUIT` like any key, and your program decides what it means (ask "Save first?", then set `QUITF`). The diag and hello examples quit immediately.

Every key restarts the console's nine-minute screen-blank timer, and turns the display back on if it had blanked.

7. Memory

Variables in slot A

Declare every variable once, as an offset from `APPRAM` (see `examples/browser/shared.a99`):

```
NFILES EQU  APPRAM+0       files found
SEL    EQU  APPRAM+2       the selected one
```

and use them with `@`: `MOV @SEL,R1`. There are 7,168 bytes between `APPRAM` and `APPTOP`; modules share state only through addresses like these, agreed in a file that every module `COPY`s (`shared.a99`). Slot A is cleared by the loader, so the first module's `INIT` can rely on zeros.

`MODRAM` (`>2200`, 256 bytes) is the one scratch area the kernel names for you, for a module that wants private work space: *it is not kept across a module switch* (and the self-test uses it for its results). For state that must survive a trip to another module use `APPRAM`.

The scratchpad (`>8300-83FF`) is the fast RAM, and the kernel and the console use nearly all of it (map in ARCHITECTURE.md). The one block that is yours is **`>8380-83BF`** (64 bytes: the loader uses it only while it copies the cartridge): good for a hot loop's counters or a small table.

Writing `>2000-3FFF` is always safe and always mapped. Never put the stack or a variable in `>A000-FFFF`: those windows are remapped.

SAMS pages for big data

The SAMS card gives you 4K pages, numbered from 0 (128 of them on a 512K card, 256 on 1M or more). The kernel and the build own some; the rest are yours:

Pages	Use
`>00-01`	never used (the console's windows)
`>02-03`	slot A

`>04-09`	**free** (24K): the CPU never maps them, so only slot B/C can show them; the loader does not test them
`>0A-0B`	the kernel
`>0C-0F`	slot C's and B's default pairs
`>10-11`	the data bank (glyphs, colour schemes)
`>12...`	your modules, two pages each, in the order `app.json` lists them (or where you say)
after the modules	`PGFREE` and up: yours, and the **page pool's** natural range

`MAPE` maps one page into window `>E000`, `MAPF` into `>F000`; afterwards `@>E000` is that page's first byte. `PAGEB` maps two *consecutive* pages into both. All of them keep `CURPB` (what slot B holds) in step, **which is what makes it safe to borrow slot B**:

```
       MOV  @CURPB,R0         what is mapped now
       .push R0
       LI   R0,MYPAGE
       BL   @MAPE
       ... use @>E000 ...
       .pop R0
       BL   @SETPB            put both windows back as they were
```

Never write a mapper register any other way (`MAPPG` does not record CURPB: it is the low-level routine). **Code never remaps the window it is running in.** Code in slot C may freely remap slot B; nothing but the kernel's main loop remaps slot C, and nothing remaps slot A. A buffer you hand to a routine must not be in `>E000-FFFF` if the routine itself remaps slot B (the rule applies to your own routines too).

The page pool

`PINIT` (R0 = first page, R1 = the page after the last) declares a pool and frees all its pages. `PALLOC` returns the lowest free page in R0 (0 when none is left); `PRELS` gives one back (twice is harmless; a page outside the pool is ignored). A page's contents are never touched. The pool is a bitmap in slot A for up to 256 pages; `FREECT` is the free count. The viewer in `examples/browser` takes pages from `PGFREE`, fills them with lines of a file, and draws from them:

```
       LI   R0,PGFREE
       LI   R1,PGFREE+8       an 8-page pool
       BL   @PINIT
       ...
       BL   @PALLOC           R0 = a page
       BL   @MAPE             map it at >E000
```

Check `NPAGES` (128 or 256) if you take pages near the top: the card the user owns may be bigger than `min_pages`, never smaller (the loader refuses it).

8. Several modules

List them in `app.json`; the build assigns pages and writes their names as equates:

```
"modules": [
  {"name": "browse", "source": "browse.a99"},
  {"name": "view",   "source": "view.a99"}
]
```

gives `PG_BROWSE` and `PG_VIEW` (the first page of each: names upper-cased) in every module that includes `kernel.equ`. `"boot"` names the module that starts (default: the first).

Switching: `GOMOD`

```
       LI   R0,PG_VIEW        the module to switch to
       MOV  @SEL,R1           an argument for its INIT
       BL   @GOMOD
```

`GOMOD` only *records* the request; the kernel switches when your `TICK` (or `INIT`) returns, so finish what you are doing first. The new module's `INIT` is called with `R0` = your page and `R1` = the argument; **slot A is unchanged, `MODRAM` is not preserved, the screen is as you left it.** `PREVPG` holds the module that ran before the most recent switch, so a module can go "back": `MOV @PREVPG,R0 / CLR R1 / BL @GOMOD`. Switching is not a call: it does not return to the old module; the old module's `INIT` runs again if you go back (so `INIT` must be able to redraw from your slot A state).

Calling: `MODCALL`

Sometimes you want a *subroutine* in another module and then to carry on here. Give the other module an extra entry (words 4 and up in its table; word 3 is reserved):

```
       AORG >C000
       DATA INIT,TICK,REDRAW,0,PING         entry 4 is PING

PING   LI   R0,>1234
       RT
```

and call it from another module with the `.callmod` macro:

```
       .callmod PG_DIAG,4        R0 = the page, R1 = the entry's address, BL @MODCALL
```

`MODCALL` maps that module into slot C, calls the routine, maps *your* module back and returns, with `R0` as the routine left it. The mapping itself changes `R0-R4` and `R7`, so pass arguments and take extra results in **`R5, R6, R8, R9`** (they cross the call untouched) or in slot A. Calls nest. The called routine runs with *its* module in slot C, so it may not touch your code or data in slot C.

Rules

- A module cannot call another's routines directly (they are not mapped): `MODCALL` or `GOMOD`.
- Shared constants and addresses go in one `shared.a99` that each module `COPY`s (see `examples/browser/shared.a99`).
- Labels in different modules are independent: two modules may both have a `LOOP1`. Only the names `kernel.equ` exports (the public routines and the equates) are reserved.
- A module that must be callable by `MODCALL` while *also* being switched to needs its entry table the same either way.

9. Files

The file service speaks to any device through the console's device service routines: the disk controller (`DSK1.`), TIPI (`PI.`), a printer port (`PIO`), and so on. It opens one file at a time.

Routine	In	Out
`FOPEN`	R0 high byte = flags; R1 = record length; R2 = the name, length-prefixed	R0 = 0 or the error
`FREAD`	R1 = buffer (up to 255 bytes)	R0 = 0, 5 at end of file, or the error; R2 = the record's length
`FWRITE`	R1 = buffer; R2 = length	R0 = 0 or the error
`FCLOSE`		R0 = 0 or the error
`FDELETE`	R2 = the name	R0 = 0 or the error
`FCNUM`	R1 = a catalog record, R2 = n (0-2)	R0 = number n of the record

Flags are added together, one from each group, and passed in the *high byte* of R0:

Group	Choices
record contents	`FDISP` (DISPLAY, default) / `FINTL` (INTERNAL)
record length	`FFIXED` (default) / `FVAR` (VARIABLE)
mode	`FIN` input, `FOUT` output (creates the file), `FUPDT` update, `FAPP` append

So a DIS/VAR 80 text file for reading is `LI R0,FIN+FVAR` with `R1 = 80`:

```
       LI   R2,NAME           NAME STRI 'DSK1.NOTES'
       LI   R0,FIN+FVAR
       LI   R1,80
       BL   @FOPEN
       MOV  R0,R0
       JNE  FAILED
LOOP   LI   R1,MYBUF          up to 255 bytes
       BL   @FREAD
       MOV  R0,R0
       JNE  DONE             5 = the end, anything else an error
       ... R2 = the line's length ...
       JMP  LOOP
DONE   BL   @FCLOSE
```

Writing is the same with `FOUT+FVAR` and `FWRITE`. **Results** (R0): 0 worked; 1-7 the DSR's error (5 beyond the end of the file, 2 bad open attributes, 7 a file error, ...); `>FF` no card has that device, or the name was longer than 53 characters. Two behaviours seen with TI's own disk controller (run in the simulator against its real DSR) that the stand-in disk does not show:

- **After the read that reports 5 (end of file) the file is already closed**; `FCLOSE` then answers 7. Treat 5 as "done" and ignore `FCLOSE`'s result after it (call it anyway: other devices may need it).
- **Opening a file that does not exist for input answers 2**, where the stand-in says 7. Do not tell "missing" from other open failures by the exact code; any non-zero result means "could not open".

The catalog. Opening a name that is only a device and a dot (`DSK1.`) as `FIN+FINTL` (INTERNAL, FIXED) with record length 38 reads the disk's catalog: the first record describes the disk, then one record per file, then an empty name. A record is a length-prefixed name, then three radix-100 numbers (`FCNUM` decodes small integers): for a file, its type (1 DIS/FIX, 2 DIS/VAR, 3 INT/FIX, 4 INT/VAR, 5 program; negative if protected), its size in sectors and its record length; for the disk, 0, its sectors and the sectors free. `examples/browser/browse.a99` shows it all.

What the service looks after for you (the rules the Lantern Coast learned the hard way): the GPL workspace, the name pointers, saving and restoring the scratchpad the DSR may change (except `>8370`, which must be what the controller left it), the controller's one-time power-up routine, the PAB in VRAM, and clearing the previous call's error bits.

What you must remember:

- A DSR call changes `R6-R9` along with `R0-R5`: keep nothing you need in them, or push it.
- It takes from milliseconds to seconds; call `KIDLE` between records if keys must not be lost (the viewer does).
- Interrupts are off during the call.
- The VRAM from `>3600` is the service's and the controller's; keep your own VRAM below it.
- One file open at a time. Open, use, close, then the next.

10. Time

The main loop runs: *wait for the frame, poll the keyboard, take one key, call `TICK`*. One pass is one frame, a sixtieth of a second (a fiftieth on a PAL console). `TICK` has the whole frame to itself: the VDP's interrupt is masked while it runs, so the console's interrupt routine (which advances the frame counter, plays sound lists and blanks the screen) does not run. If `TICK` takes longer than a frame, the lost frames are not made up: the next pass starts at once. `FRAME` counts main-loop passes (and the frames `KIDLE` sees), so it is a *tick* count, not exactly time.

Long work. If a task takes more than a few frames (reading a file, sorting, drawing a page), poll the keyboard between steps so typed keys are not lost:

- `KIDLE` checks the VDP's frame flag and, if a frame has passed, polls. Call it between VDP transfers, never in the middle of one. It uses `R0-R7` and `R12`.
- Inside a tight byte loop, read `VDPSTA` yourself and call `KPFR` when its top bit is set; `KPFR` keeps every register but `R0`.

Keys land in the buffer; take them with `KEYGET` when the work is done (or *flush* them if the user should not be able to queue them).

Frame timing. `VSYNC` waits for the next frame and is the only place interrupts are allowed. To wait N frames: loop `BL @VSYNC` N times (or `BL @KWAIT` for a key).

Sound. The console's interrupt routine plays *sound lists* (the console's own sound format) from VDP RAM or GROM, one block per frame: point `SNDPTC` (`>83CC`) at the list, set bit 0 of `SNDFLG` (`>83FD`) for a list in VRAM, and `SNDDUR` (`>83CE`) to 1 to start it. To silence everything write the four attenuation bytes `>9F >BF >DF >FF` to `SOUND` (as `KEXIT` does). The kernel has no

further sound API in 1.0, and the sound-list path has not been exercised by the tests.

Screen blank. After about nine idle minutes the console blanks the display (it writes VR1 from `VR1CPY` with the display bit cleared); any key restores it (`KPUT` writes `VR1CPY` back with the display bit set). If you change video mode, keep `VR1CPY` (`>83D4`) equal to the VR1 value that mode runs with (Text 80: `>F0`), display bit set, or the screen will come back in the wrong mode.

11. Quitting

Set `QUITF` non-zero (`SETO @QUITF`) from `TICK` or `INIT`. After the routine returns the kernel (`KEXIT`) silences the sound chip, waits up to a second for the keys to be released (so the console does not see the same FCTN = and quit its own menu at once), restores the F18A to its power-up state (30 rows, position attributes, the stop-sprite register, the 16 standard colours), switches the SAMS mapping off, selects cartridge bank 0 (so the console's menu finds your program) and resets the console, exactly as FCTN QUIT does. Close any open file first; there is no exit hook.

12. Building

`python3 tools/lkbuild.py APPDIR` reads `APPDIR/app.json`:

```
        {
          "name": "MY APP",             menu name and refusal screens (20 characters)
          "min_pages": 128,             SAMS pages required: 32, 64, 128 or 256
          "modules": [
            {"name": "main",  "source": "main.a99"},
            {"name": "other", "source": "other.a99", "page": "0x20"}
          ],
          "boot": "main",
          "data": [ {"file": "words.bin", "page": "0x40"} ]
        }
```

- `min_pages` is what your program needs; a smaller card is refused on a screen that names the app and the size ("NEEDS A SAMS MEMORY CARD WITH 512K").
- **Module pages** are assigned from `>12`, two each, unless you give `"page"` (an even start is conventional; allowed: `>04-09` and `>12` up). Overlaps, pages past `min_pages` and the kernel's own pages are errors.
- `data` files are copied into SAMS pages at start-up, in 8K pieces to consecutive page pairs: a dictionary, a map, a table. The equate `PD_<FILE>` gives the first page. The image grows with them (8K per piece), so the cartridge's size is the limit; Press's 512K image has run in js99er but, at the time of writing, not on a real FinalGROM 99.
- `PGFREE` (equate) is the first page the build leaves unassigned: where your pool starts.
- The output `build/APPDIR/NAME8.bin` is the cartridge image: 8K banks, padded to a power of two, bank 0 the loader. A real FinalGROM 99 and js99er take it as is.
- **Listings** (`.lst`) of every bank are kept beside it; `kernel.equ` is the symbol table.
- **Your own font and colour schemes:** put a `databank.a99` in your application's directory (copy `kernel/databank.a99` and edit): the build looks there first. Keep its layout: 256 glyphs at `>E000`, five schemes of 16 words at `>E800`.

The build runs `tools/font.py` and `tools/keytab.py` first (they write `gen/font.a99` and `gen/keytab.a99`: generated, never edit them).

13. Testing

You can run and test an application without a TI.

```
python3 tools/sim9900.py build/myapp/myapp8.bin --keys "<ALPHA>hello<ENTER>" --frames 300
```

prints the screen as text. In Python (see `tests/test_kernel.py`):

```
        import sim9900
        cart = open('build/myapp/myapp8.bin', 'rb').read()
        m, cpu, halt = sim9900.run(cart, keys='<C-t><WAIT 600>', frames=1000)
        print(sim9900.screen(m))                # 30 lines of 80 characters
        m.rw(0x2300)                             # a word of memory (a variable)
        m.vram[0x1000:0x1008]                    # VRAM
        m.palette, m.vregs                       # the F18A's palette and registers
```

Key scripts: plain characters type themselves (capitals with SHIFT), `<ENTER>`, `<SPACE>`, `<C-x>` CTRL + x, `<F-x>` FCTN + x (`<F-=>` quit, `<F-e>` up, `<F-x>` down, `<F-s>` `<F-d>` left and right), `<S-x>` SHIFT, `<ALPHA>` toggles ALPHA LOCK (**it starts down**,

as in js99er, so type `<ALPHA>` first for lower case), `<WAIT n>` waits n frames, `<HOLD n>` makes the next key a hold of n frames. The first key goes down at frame 60.

Pictures. `tools/f18render.py` draws any F18A mode from the simulator's state: `f18render.render(m)` returns a numpy picture (256 x 192, 240 rows, or 512 wide in Text 80) and `f18render.save(m, 'shot.png')` writes it; compare a pixel with `f18render.palette_rgb(m)[entry]`. `tests/test_kernel.py modes` does that for every screen of `examples/modes`.

Hooks and disks. Pass `setup=fn(m)` to prepare the machine before it runs: `m.dvfiles['NAME'] = [b'record', ...]` puts a file on the simulator's DSK1 (all DIS/VAR 80 unless `m.ftypes['NAME'] = (variable?, internal?, record length)` says otherwise), `m.bad_pages.add(0x2A)` makes a page fail its test, `run(..., pages=64)`, `sams=False`, `f18a=False` try other machines. After a quit, `m.at_reset` holds the F18A registers, palette and mapping at the console reset.

Test what a person would see: read the whole screen, not a line; give a run enough frames to settle; check the cursor and the data, not only the first key. `tests/test_kernel.py` is the model (the groups `boot refusal ram keys quit modules files browser rom`).

The simulator is not the machine. It models the TMS9900, the VDP and the F18A's text modes and DMA, the SAMS card, the keyboard matrix and a stand-in disk controller. It does not model timing, and its stand-in disk is kinder than a real controller (see the two differences in part 9). **Real disk controller:** with `TIFDC_ROM=path` to the controller's 8K ROM (join the two 4K EPROM images, low half first) tests can give the machine a `tifdc.TIFDC(image=...)` in `setup` (see the `realdsr` group): TI's own DSR then runs on an emulated WD1771 and a 90K disk image, as in js99er. Test on js99er and on real hardware before you believe it (part 14). **Real console ROM:** if you have your own copy of the TI console ROM (8K, raw or as an EA5 image) set `TI_CONSOLE_ROM=path` (or pass `rom=`): the simulator then runs TI's real interrupt routine instead of a model of it, which catches mistakes in how a program treats the console's variables. The ROM is TI's: it is never part of this repository; keep it outside.

The self-test on hardware. `examples/diag` is meant to be the first thing run on a new machine: press CTRL T for the kernel's self-test (hardware, paging, pool, module call, keyboard, screen, DMA, a DSK1 write/read/delete), CTRL 3 to cycle colour schemes, every key shows its code.

14. The checklist: what has already cost a day

1. **Registers across calls.** Check each callee's "Uses" (`PUTS` R0-R7, `DECW` R0-R8, `KIDLE` R0-R7 and R12, a DSR call R0-R9). Keep loop state in memory.
2. `.leave` **clobbers the flags.** Test a result register after the call.
3. `R0` **is not an index register;** `MOVB` **leaves the low byte;** `SLA Rn,0` **shifts by R0.**
4. `EVEN` **after byte data** (or a code label lands on an odd address); count `TEXT` lengths or use `STRI`.
5. **Jumps reach about 128 words**: when one goes out of range, invert the condition around a `B @label`.
6. **Expressions evaluate left to right**: parenthesise.
7. **Code never remaps the window it is running in**; slot C belongs to the kernel's main loop.
8. **Map slot B only with** `MAPE`/`MAPF`/`PAGEB`/`SETPB`, so `CURPB` is true; save and restore it when you borrow.
9. **Interrupts are masked outside** `VSYNC`**.** Do not `LIMI 2` in your code; do not poll the keyboard in the middle of a VDP transfer.
10. **Never write an F18A register you do not need**, and do not unlock the F18A twice (the second unlock locks it): the kernel handled both.
11. **Power-up RAM is random**: clear what you use. (The loader clears slot A; nothing else is cleared.)
12. **A DSR call changes R6-R9 and takes time**; keep the VRAM from `>3600` for the file service.
13. **The 9918 is not the F18A**: the F18A's extras (30 rows, per-cell colour, palette, enhanced colour, layers) are what the kernel gives you; do not rely on a plain TI's video. `VMODE` **rewrites the screen**, and the attribute byte means roles in the normal colour modes and flags in the enhanced ones; **Graphics II and file I/O do not mix** (they share VRAM).
14. **The console's frame counter needs** `>83FC`**'s high byte non-zero** (the kernel sets it at start and after every DSR call). Do not clear it.
15. **Slot A is 7K.** If you need more, it is a SAMS page.
16. **It works in the simulator** is the beginning of a test, not the end: js99er, then the real machine.

15. What the kernel does not do (version 1.1)

- Video: every F18A mode is reachable (`VMODE`, `VREG`), but the kernel only *draws* for you in the glyph modes (Text 80/40, Graphics I); sprites, Graphics II, Multicolor and the layers you write to VRAM yourself. No sprite library, no scroller, no tile editor. A module that changes registers beyond `VMODE`'s must keep `VR1CPY` right.
- No sound API beyond the console's sound lists.

- No heap, no text-editing engine, no menus: those are applications (Press has an editor built on the ancestor of this kernel; see FROM-PRESS.md).
- One file open at a time; no relative-record files beyond what the PAB offers, no file-size query.
- Joysticks are not read (the keyboard scan covers the keyboard only).
- Cards over 1M are used as 1M.
- No cooperative multitasking: one module's `TICK` runs at a time.

Appendix: a second module in five minutes

1. `app.json`: add `{"name": "about", "source": "about.a99"}`.
2. `about.a99`: copy the scaffold; give it its own `INIT` that draws and `TICK` that, on any key, does `MOV @PREVPG,R0 / CLR R1 / BL @GOMOD`.
3. In `main.a99`'s `TICK`: on `CI R0,>8100` (CTRL A) do `LI R0,PG_ABOUT / CLR R1 / BL @GOMOD`.
4. Build and run. In the first module, make `INIT` redraw everything from slot A state, because it runs again when you come back.